Modules – Config

Menu: Modules > Config

Configures the hardware settings of the connected module.

Module Identity

The underlined line at the top shows which module the module pages are editing — the module number, the board's description (or reported label), and its connection state. It updates live while the page is open. If it turns red with **"Two boards are answering as module N!"**, two powered boards share the same module ID — disconnect one and give it its own ID before continuing.

If the module's board needs a firmware update, the line turns red and reads **Module N (Firmware update required)**. The module is not used, and nothing is sent to it, until its firmware is updated.

Below it, while the module is connected, is the board type it reports, e.g. **Teensy Rate** or **ESP Rate**. Nothing is shown while it is not connected.

Boards are given their module number and description on the **Modules > Boards** page, which also lists new boards.

Setting	Description
Module to edit	The module to edit, listed as *N (Connected)*, *N (Not connected)* or *N (Update firmware)*. All module pages (Config, Communication, Pins, Relay Pins, Valves) and the Send button work on this module; other connected modules ignore the settings. A new choice takes effect straight away, without Save. It is refused while any module page has unsaved changes.

Load Preset

Loads a full starting configuration for a known board — the settings on this page, the sensor pins and the relay pins — and saves it. It uses the type of the board reporting on this module. When no board is reporting, it asks which type to load.

Preset	Microcontroller
Nano (RC12-3)	Arduino Nano
Teensy (RC11-2)	Teensy 4.1
ESP32 (RC15)	ESP32

After loading, check the Pins, Relay Pins and Valves pages against your own board, then press **Send**. Load preset is refused while any module page has unsaved changes.

> **Note:** A preset replaces the module's pin and relay settings. Only use it during initial setup or when changing hardware. It does not change the module number.

Sending to the Module

Press **Send** to give the module its settings. Send is refused, with a message, when no board is connected on the module being edited, or when its board needs a firmware update. Saved but unsent changes on the Boards page are dropped when that happens; save them again when the board is ready.

1. Relay Type

Value	Description
0	No relays
1	GPIOs — use micro-controller pins directly
2	PCA9555 — 8-relay module
3	PCA9555 — 16-relay module
4	MCP23017
5	PCA9685
6	PCF8574

2. Pin Assignments by Board

Parameter	RC5 (Nano)	RC8 (Nano)	RC11 (Teensy)	RC11-2 (Teensy)	RC12 (Nano)	RC12-3 (Nano)	RC15 (ESP32)
Sensor Count	2	2	2	2	1	2	2
Relay Control	4	4	1	1	2	4	5
Flow 1	2	2	28	28	D3	3	17
Flow 2	3	3	29	29	—	2	16
DIR 1	4	4	37	37	D6	4	32
DIR 2	6	6	14	14	—	6	25
PWM 1	5	5	36	36	D9	5	33
PWM 2	9	9	15	15	—	9	26
Work Pin	—	—	—	30	—	15	—
Pressure Pin	—	—	—	40	—	14	—

3. Nano Pin Reference

Pin Name	Number
D0–D13	0–13
A0–A7	14–21

Controls

Control	Action
Load preset	Load and save the starting configuration for the board type (see above)
Rescan	Reset the settings on this page only. Save or Cancel as for any change
⊗ Cancel	Discard changes
Save	Save settings

Modules – Pins

Menu: Modules > Pins

Manually assigns the GPIO pins used for sensors and motor/valve control on the module. The grid shows one row per sensor (the sensor count is set in Modules > Config). Pin values range from 0–50. A blank value (displayed as "–") means the pin is not assigned.

Grid Column	Description
Sensor	Sensor/product number (read-only)
Flow	Input pin for the flow sensor
Dir	Output pin for the motor/valve direction channel
PWM	PWM output pin for the motor/valve
Bin	Input pin for the bin-empty sensor. Leave unassigned ("–") if the product has no bin sensor — the bin alarm is off for that product.
Invert	Inverts the bin sensor reading. Unchecked: an open input reads *empty* (sensor closes to ground while material is present). Checked: a grounded input reads *empty*.

Setting	Description
Work Pin	Input pin for the work/implement switch
Momentary	Enables momentary switch behavior on the work pin
Invert Work Switch	Inverts the work switch reading to suit the sensor type. Unchecked: an open input reads as *working* (normally-closed sensor). Checked: a grounded input reads as *working* (normally-open sensor). In momentary mode the toggle occurs on the opposite edge. Older module firmware ignores this setting.
Pressure Pin	Analog input pin for the pressure sensor

Press **Rescan** to clear all pin assignments. Duplicate pin numbers are rejected on Save.

> **Tip: Load preset** in Modules > Config fills in these pin values for a standard board. Only change them if your hardware differs from the standard layout.

Bin-Empty Alarm

When a Bin pin is assigned, the module watches that input and reports the bin state to RateController. The bin sensor input uses an internal pull-up, so a simple switch or level sensor wired between the pin and ground is all that is required. The module filters the reading — the sensor must hold its state for about 2.5 seconds before the reported state changes, so brief flickers from material sloshing are ignored.

- The alarm is only raised **while the master switch is on** — an empty bin at start-up does not alarm until you begin applying.
- When it triggers, the alarm button shows **Bin Empty** with the product name, the product's tile turns red, and on the selected product the quantity caption changes to **Bin Empty** with the quantity value shown as a red 0.
- The displayed 0 is display-only — the recorded tank amount and quantity applied are not changed, and the real value returns as soon as the alarm clears.
- The alarm clears immediately when the bin refills or the master switch is turned off. Tap the alarm button to silence the buzzer.

> **Tip:** Use a plain digital pin for the bin sensor. Inputs with analog conditioning on the board (such as the pressure input) load the internal pull-up and may not read reliably as a digital input.

Controls

Control	Action
Cancel	Discard changes
Save	Save settings

Modules – Relay Pins

Menu: Modules > Relay Pins

Assigns the physical GPIO pins to each of the 16 relay outputs on the module. Pin values range from 0–50. A value of 255 (displayed as "–") means the relay pin is not assigned.

- **Low Relays (1–8):** Labeled "Onboard Relays" or "Remote Relays" depending on module config.
- **High Relays (9–16):** Labeled "Onboard Relays" or "Remote Relays" depending on module config.

Press **Rescan** to clear all relay pin assignments.

> **Tip: Load preset** in Modules > Config fills in the relay pins for a standard board.

Controls

Control	Action
Cancel	Discard changes
Save	Save settings

Modules – Valves

Menu: Modules > Valves

Selects the wiring mode for the valve or motor driver.

Option	Description
2-Wire	Motorized or latching valve — the driver actively powers the valve in both directions to open and close it. 
3-Wire	One switched signal wire, one power wire and one ground wire. Compatible with standard solenoid valves and motorized ball valves that manage open/close internally. Default for most installations. 

Select the mode that matches how your valve or motor driver is wired to the module.

Controls

Control	Action
Cancel	Discard changes
Save	Save settings